DEEPLENSE

GSoC 2026 Evaluation Test Blueprint

Author: Rastin Aghighi
Organization: ML4SCI — Machine Learning for Science
Project: Unsupervised Super-Resolution and Analysis of Real Lensing Images
Mentors: Michael Toomey (MIT), Pranath Reddy, Sergei Gleyzer
Deadline: April 1, 2026, 18:00 UTC

Execution Pipeline

Table of Contents

1. Dataset Specifications
2. EDSR-Baseline Architecture
3. Composite Loss Function
4. Task VI.A — Simulated Super-Resolution
5. Task VI.B — Real Telescope Super-Resolution
6. Task I — Multi-Class Classification
7. Evaluation Protocol
8. Visualization Suite
9. Submission Checklist

1. Dataset Specifications

1.1 Task VI.A — Simulated Strong Lensing (No Substructure)

Property	HR Images	LR Images
Path	Dataset/Dataset/HR/	Dataset/Dataset/LR/
Count	10,000	4,492
Matched Pairs	4,492	4,492
Naming	sample{N}.npy	sample{N}.npy
Shape	(1, 150, 150)	(1, 75, 75)
Dtype	float64	float64
Value Range	[0.0000, 1.0000]	[0.0011, 1.0088]
Scale Factor	2× (75 → 150)	
Split	90:10 → ~4,042 train / ~450 test	

1.2 Task VI.B — Real Telescope (HSC/HST Pairs)

Property	HR (HST)	LR (HSC)
Path	Dataset 3B/Dataset/HR/	Dataset 3B/Dataset/LR/
Count	300	300
Naming	HR_{N}.npy	LR_{N}.npy
Shape	(1, 128, 128)	(1, 64, 64)
Dtype	float32	float32
Value Range	[0.0019, 1.0108]	[-0.0044, 0.9987]
Scale Factor	2× (64 → 128)	
Split	90:10 → 270 train / 30 test	

Note: Values outside [0,1] are physically meaningful — negatives from sky subtraction, >1.0 from bright source peaks. Data is **never clipped during training**.

1.3 Task I — Classification (Pending Download)

Three classes: no substructure, subhalo substructure, vortex substructure. Dataset not yet downloaded — separate Google Drive link from evaluation page. Expected: .npy single-channel images, min-max normalized.

Exact dimensions TBD.

2. EDSR-Baseline Architecture

We use EDSR-baseline (Lim et al., 2017) — 16 residual blocks, 64 feature channels, no batch normalization, 2× upscaling via sub-pixel convolution. The key insight from the original paper is that removing batch normalization improves super-resolution performance: BN normalizes activations to zero mean and unit variance, which constrains the network's output range — counterproductive for pixel-value regression where we need to reconstruct precise intensity values spanning the full dynamic range of astronomical images.

Full Architecture

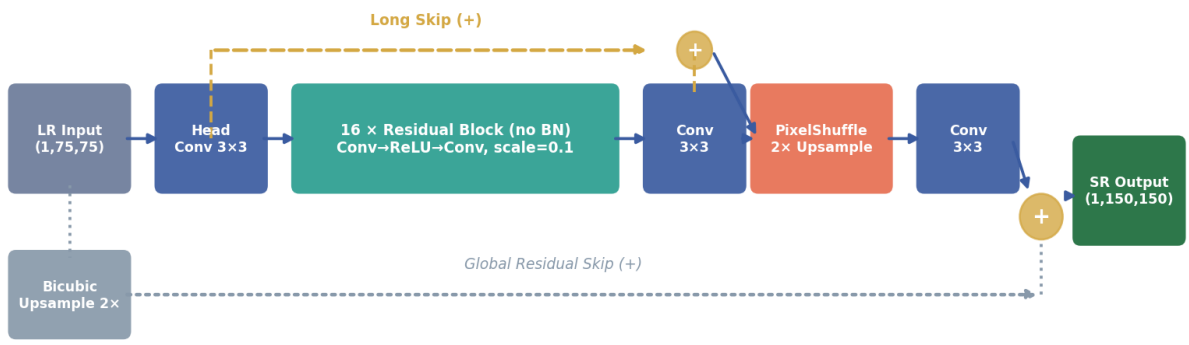

Figure 1: EDSR-baseline architecture with global residual learning.

Residual Block Detail

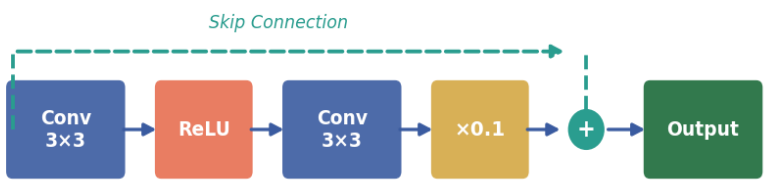

Figure 2: Single residual block — no BN, residual scaling = 0.1.

Key Design Decisions

- **No Batch Normalization:** BN normalizes features to zero mean/unit variance, limiting range flexibility for pixel regression. Removing it improved PSNR by ~0.1–0.3 dB in the original EDSR paper.
- **Residual Scaling (0.1):** Multiplies each residual block's output by 0.1 before adding to the skip, preventing gradient explosion in deep residual chains.
- **Global Residual Learning:** Output = Network(LR) + Bicubic(LR). The model only learns the high-frequency residual, making optimization easier and convergence faster.
- **Sub-pixel Convolution (PixelShuffle):** Rearranges a $(C \cdot r^2, H, W)$ tensor into (C, rH, rW). More efficient than transposed convolution and avoids checkerboard artifacts.

- **Fully Convolutional:** Accepts any input spatial resolution. Critical for transfer from VI.A (75×75 input) to VI.B (64×64 input) without architecture changes.

Component	Parameters	Notes
Head (Conv 3×3)	~640	1→64 channels
Body (16 ResBlocks)	~1,180,672	64 channels, 2 convs each
Body Final Conv	~36,928	64→64
Tail (PixelShuffle + Conv)	~114,945	64→256→64→1
Total	**~1.37M**	

3. Composite Loss Function

We use a physics-informed composite loss that combines pixel-level reconstruction accuracy with physical constraints specific to gravitational lensing:

$$\mathbf{L}_{total} = \mathbf{L}_{L1} + \lambda_{flux} \cdot \mathbf{L}_{flux} + \lambda_{bp} \cdot \mathbf{L}_{bp}$$

Figure 3: Composite loss function dataflow.

Component 1: L1 Reconstruction Loss

$$\mathbf{L}_{L1} = (1/N) \Sigma |SR_i - HR_i|$$

Pixel-wise mean absolute error. Preferred over L2/MSE because L2 heavily penalizes outliers, causing the model to predict the conditional mean — resulting in blurry outputs. L1 predicts the conditional median, producing sharper reconstructions.

Component 2: Flux Consistency Loss

$$\mathbf{L}_{flux} = |\Sigma SR_i - \Sigma HR_i| / N_{pixels}$$

Penalizes differences in total integrated intensity (flux) between SR output and HR ground truth. This is physically motivated: gravitational lensing conserves total photon flux — the lens redistributes light across the image but does not create or destroy photons. $\lambda_{flux} = 0.05$ (small weight because the flux sum aggregates over all pixels).

Component 3: Back-Projection Loss

$$\mathbf{L}_{bp} = (1/N) \Sigma |Downsample(SR)_i - LR_i|$$

Ensures the SR output, when downsampled back to LR resolution, reproduces the original low-resolution observation. This constrains the solution space to physically plausible reconstructions that are consistent with what was actually observed. The downsampling uses bicubic interpolation (matching the assumed degradation model). $\lambda_{bp} = 0.1$ (gentle consistency check — too high would force the model toward bicubic output).

Key insight for GSoC proposal: The back-projection loss only requires the degradation model (HR→LR mapping), not paired data. This makes it directly applicable to unsupervised SR.

4. Task VI.A — Simulated Super-Resolution

4.1 Training Configuration

Parameter	Primary (Composite)	Ablation (L1-only)
Optimizer	Adam	Adam
Learning Rate	1e-4	1e-4
Scheduler	ReduceLROnPlateau(0.5, pat=5)	ReduceLROnPlateau(0.5, pat=5)
Early Stopping	patience=10	patience=10
Max Epochs	100	100
Batch Size	16	16
Loss	L1 + 0.05·Flux + 0.1·BP	L1 only
Augmentation	90° rot + h-flip + v-flip	90° rot + h-flip + v-flip
Seed	42	42
Save Path	weights/edsr_sim_best.pth	weights/edsr_sim_l1only.pth

4.2 Data Augmentation

Gravitational lensing images have no preferred orientation on the sky, making geometric augmentations physically motivated rather than just a regularization trick:

- **Random 90° rotation** ($k \in \{0,1,2,3\}$): Applied identically to HR and LR. No continuous rotation — would introduce interpolation artifacts in ground truth.
- **Random horizontal flip** (p=0.5): No preferred handedness in lensing.
- **Random vertical flip** (p=0.5): Same reasoning.
- **Effective multiplier:** Up to 8× dataset size (4 rotations × 2 flip states).
- **Not used:** Color jitter (single-channel), elastic deformation (distorts physics), cutout/mixup (inappropriate for scientific imaging).

4.3 Self-Ensemble Inference (EDSR+)

At test time, run the model on all 8 augmented versions of each LR input (4 rotations × 2 flip states). De-augment each SR output (reverse flip, reverse rotation), then average all 8 outputs. This suppresses directional artifacts learned during training and typically adds +0.1 to +0.3 dB PSNR for free — no retraining needed. Especially effective for astronomical images where all 8 orientations are equally valid.

4.4 Ablation Study

Train two identical models — one with composite loss, one with L1 only — using the same seed, split, augmentation, and optimizer. Compare on all metrics. Expected result: composite loss should show meaningfully lower flux error (validating the physics-informed approach) with similar or slightly better PSNR/SSIM.

Method	PSNR ↑	SSIM ↑	MSE ↓	Flux Err ↓
Bicubic	—	—	—	—
L1-only EDSR+	—	—	—	—
Composite EDSR+	—	—	—	—

Table: VI.A ablation results (to be filled after training).

5. Task VI.B — Real Telescope Super-Resolution

The core challenge: adapt a model trained on clean simulated data to real telescope observations with only 300 paired images (270 train / 30 test). The domain gap includes different PSFs, noise characteristics, pixel scales, background sources, and cosmic ray artifacts.

5.1 Transfer Learning: 3-Stage Gradual Unfreezing

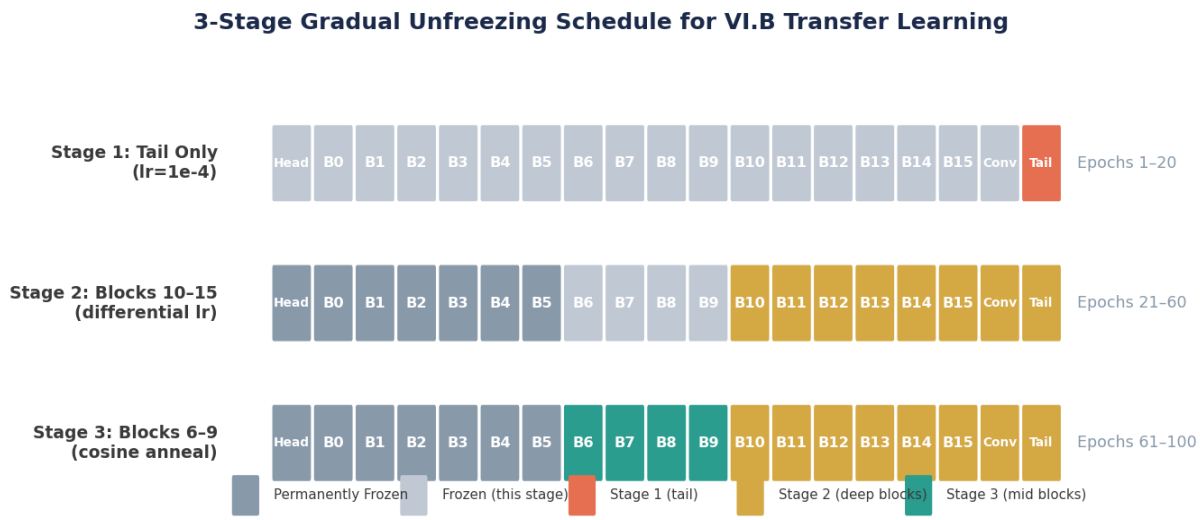

Figure 4: Gradual unfreezing schedule. Blocks 0–5 remain permanently frozen.

Parameter	Stage 1	Stage 2	Stage 3
Epochs	1–20	21–60	61–100
Trainable	Tail only	Tail + Blocks 10–15	Tail + Blocks 6–15
Frozen	Head + Body	Head + Blocks 0–9	Head + Blocks 0–5
Tail LR	1e-4	5e-5	1e-5
Deep blocks LR	—	1e-5 to 5e-6	5e-6
Mid blocks LR	—	—	1e-6
Scheduler	ReduceLR(pat=3)	ReduceLR(pat=5)	CosineAnneal(T=40)
Early stop	patience=5	patience=8	patience=10
Regularization	L2 decay (1e-4)	L2-SP (α=0.01)	L2-SP (α=0.01)

5.2 L2-SP Regularization

$$L_{L2\text{-}SP} = \alpha \cdot \Sigma (\theta_i - \theta_i^{pretrained})^2$$

L2-SP (Li et al., 2018) penalizes parameters for deviating from their pretrained values, not for magnitude. This prevents catastrophic forgetting of useful features learned on simulated data while still allowing domain adaptation. $\alpha = 0.01$. Applied in Stages 2 and 3 only (Stage 1 adapts the tail, which was not pretrained for this pixel scale).

5.3 Ablation: Fine-tuned vs From-Scratch

Train a second EDSR from random initialization on the same 270 real images (same loss, augmentation, optimizer, seed). Expected result: from-scratch significantly underperforms, validating transfer learning. With only 270 images, the from-scratch model will likely overfit before learning meaningful features.

Method	PSNR ↑	SSIM ↑	MSE ↓	Flux Err ↓
Bicubic	—	—	—	—
From-scratch EDSR+	—	—	—	—
Fine-tuned EDSR+	—	—	—	—

Table: VI.B ablation results (to be filled after training).

6. Task I — Multi-Class Classification

Binary and multi-class classification of gravitational lensing images into three morphological categories: no substructure, subhalo substructure, and vortex substructure.

6.1 Model: ResNet-18

torchvision ResNet-18 pretrained on ImageNet. Modification: repeat single-channel input 3× along channel dimension (preserves all pretrained weights). Replace final FC layer: Linear(512, 3). Alexander et al. (2020) achieved AUC 0.984 with this architecture on lensing data.

Parameter	Value
Optimizer	Adam, lr=1e-4
Loss	CrossEntropyLoss
Scheduler	ReduceLROnPlateau(factor=0.5, patience=3)
Early Stopping	patience=7
Max Epochs	50
Batch Size	32
Augmentation	Continuous rotation (0–360°), h-flip, v-flip
Split	90:10
Metrics	ROC curve (per class, OvR), AUC (per class + macro)

7. Evaluation Protocol

7.1 Two-Track Evaluation (Super-Resolution)

Track 1 — Engineering Metrics (clipped to [0, 1]):

$$\textbf{MSE} = (1/N)\ \Sigma\ (SR_i - HR_i)^2 \quad \text{where } SR, HR \in [0, 1]$$

$$\textbf{PSNR} = 10 \cdot \log_{10}(MAX_I^2 / MSE) \quad \text{where } MAX_I = 1.0$$

SSIM = structural similarity with data_range = 1.0

Track 2 — Physics Metrics (raw, unclipped):

$$\textbf{Flux Error} = |\Sigma\ SR_{raw} - \Sigma\ HR_{raw}| \ / \ |\Sigma\ HR_{raw}|$$

7.2 Statistical Reporting

- **Mean ± standard deviation** across all test images
- **Median** (outlier-resistant central tendency)
- **Bootstrap 95% confidence interval** (1000 resamples with replacement)
- **Per-image distribution** (violin plots)

7.3 Reference Benchmarks

	Bicubic (est.)	Our Target	DiffLense
VI.A PSNR	24–27 dB	28–33 dB	—
VI.A SSIM	0.70–0.80	0.85–0.95	0.984
VI.B PSNR	22–25 dB	25–30 dB	35.07
VI.B SSIM	0.60–0.75	0.75–0.88	0.839

8. Visualization Suite

Every notebook includes a comprehensive set of visualizations. These are what mentors look at most carefully — they reveal whether you understand the data and model behavior, not just whether the code runs.

8.1 Super-Resolution Notebooks (VI.A and VI.B)

#	Visualization	Details
1	Sample pairs	5 HR/LR pairs, log-normalized, inferno colormap
2	Training curves	Dual panel: loss components + PSNR/SSIM dual-axis
3	Visual comparison	N×4 grid: LR / EDSR / EDSR+ / HR, shared normalization
4	Error maps	3×5 grid: |bicubic-HR| / |EDSR-HR| / |EDSR+-HR|, hot colormap
5	Metric distributions	3 violin plots: PSNR, SSIM, Flux Error per method
6	Failure analysis	5 worst PSNR: LR / SR / error map with annotations
7	Ablation table	Formatted comparison: composite vs L1 (or fine-tuned vs scratch)

VI.B also includes:

- **Domain comparison:** Side-by-side simulated vs real images showing the domain gap
- **Stage-wise training curves:** Combined loss/PSNR across all 3 stages with vertical transition markers

8.2 Classification Notebook (Task I)

#	Visualization	Details
1	Class distribution	Bar chart of samples per class
2	Sample images	3×5 grid (3 classes × 5 samples)
3	Training curves	Loss + accuracy (train and val)
4	Confusion matrix	Normalized heatmap
5	ROC curves	Per-class OvR + macro-average, AUC in legend
6	Correct predictions	2 per class with confidence scores
7	Misclassifications	Worst confidence errors with true/predicted labels

9. Submission Checklist

GitHub Repository

- README.md — project overview, setup instructions, results summary
- src/ — shared utility modules (edsr.py, losses.py, dataset.py, metrics.py, visualization.py)
- notebooks/ — all 3 .ipynb files (VI.A, VI.B, Task I)
- weights/ — all .pth model weight files
- figures/ — all saved plots
- requirements.txt — torch, torchvision, numpy, matplotlib, scikit-image, scikit-learn

Per Notebook

- Runs top-to-bottom without errors
- All markdown cells explain reasoning, not just what the code does
- All figures saved to figures/ directory
- Model weights saved to weights/ directory
- Evaluation metrics printed clearly
- Discussion section at the end

Google Form Submission

- Notebook files (.ipynb)
- GitHub repo link
- Trained model weight files (.pth)

Quality Checks

- No hardcoded absolute paths (relative from project root)
- Reproducible: seed=42 set everywhere (random, numpy, torch, cuda)
- Clean code, consistent style, meaningful variable names
- No unused imports or dead code
- All plots have titles, axis labels, legends where appropriate